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3-D Global Hybrid Modeling of the Martian Plasma Environment (GCPIC-M): Impact of IMF Cone Angle on Ion Escape

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Key Points:

- The GCPIC-M model has been developed for three-dimensional global simulations of the Martian plasma environment
- Foreshock and waves regulate planetary ion transport, in addition to tailward escape and plume outflow
- The Martian ion escape rate varies with IMF cone angle θ and reaches a maximum around θ = 10°, where strong foreshock wave activity occurs

Supporting Information:

Supporting Information may be found in the online version of this article.

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Abstract Mars possesses a unique plasma environment formed by the interaction between the solar wind and the Martian induced magnetosphere. Ion escape from Mars has long been recognized as a key process governing the evolution of the Martian atmosphere and climate. Global hybrid simulations provide a powerful tool for investigating the large-scale Martian plasma environment while retaining ion kinetic effects. However, many recent hybrid models simplify ionospheric ion production by prescribing ion emission from the ionosphere. In this study, we have developed a new three-dimensional (3-D) global hybrid simulation model, the general curvilinear plasma simulation code for Mars (GCPIC-M), in which ionospheric ion production is self-consistently determined through photochemical reactions. Under given upstream solar wind conditions, the simulated heavy ion density profiles in the ionosphere agree well with Mars Atmosphere and Volatile EvolutioN (MAVEN) observations. Using this model, we have quantified planetary ion escape and investigated its dependence on the interplanetary magnetic field (IMF) cone angle. The simulated escape rates are on the order of 10^{24} s^{-1}, consistent with typical values in observations. Moreover, the escape rate exhibits a nonmonotonic dependence on IMF cone angle, reaching a maximum of $\sim 7 \times 10^{24}$ s^{-1} near ∼10°, where foreshock wave activity is enhanced. These results highlight the kinetic nature of Martian ion escape and the importance of IMF geometry. By self-consistently linking ion production, transport, and atmospheric escape, GCPIC-M provides a unified framework for investigating ion kinetic processes in the Martian plasma environment.

Plain Language Summary Mars has undergone a dramatic climate transition from a warm and wet environment to the cold and tenuous atmosphere observed today. Atmospheric ion escape is thought to play an important role in this long-term evolution. In this study, we have developed a three-dimensional global simulation model of the Martian plasma environment, in which ionospheric ions are self-consistently produced through photochemical reactions. Under given upstream solar wind conditions, the simulated ionospheric density profiles show good agreement with observations from the Mars Atmosphere and Volatile EvolutioN (MAVEN). We have further quantified the ion escape rate and examined its dependence on the interplanetary magnetic field cone angle, defined as the angle between the solar wind flow and interplanetary magnetic field directions. The escape rate reaches a maximum near a cone angle of ∼10°, corresponding to strong foreshock wave activity upstream of the Martian bow shock. These results provide new insight into plasma processes and atmospheric escape on Mars.

1. Introduction

Mars is surrounded by a plasma environment shaped by the interaction between the solar wind, the Martian atmosphere, and localized crustal magnetic fields (Acuña et al., 1998; Brain, 2006; Fang et al., 2015; Nagy et al., 2004). This interaction gives rise to distinct plasma regions, including a bow shock, a magnetosheath, the induced magnetospheric boundary (IMB), the ionosphere, and the magnetic tail (Bertucci et al., 2005; Gruesbeck et al., 2018; Nagy et al., 2004; Trotignon et al., 2006). Among these regions, the Martian ionosphere interacts directly with the solar wind and serves as the primary source of planetary ions, with major ion species including H^+, O^+, O_2^+ and CO_2^+ (Barabash et al., 2007; Fox, 2009).

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Geological and geomorphological evidence suggests that early Mars once possessed abundant liquid water and a denser atmosphere, implying a much warmer and wetter climate in the past (Goudge et al., 2017; Hynek et al., 2010; Mustard et al., 2008). The subsequent loss of atmospheric ions through solar wind-driven escape is widely regarded as a key factor in the long-term evolution of the Martian climate (Barabash et al., 2007; Fang et al., 2010; Jakosky et al., 2018; Lammer et al., 2008). Understanding the Martian plasma environment is therefore essential for investigating atmospheric escape and the evolution of Mars from a potentially habitable world to the cold and dry planet observed today.

Previous studies have shown that ion escape from Mars mainly occurs through two major channels: tailward escape and plume escape (Curry et al., 2015; Y. Dong et al., 2017; Inui et al., 2019; Nilsson et al., 2023; Shen et al., 2024). Tailward escape is primarily associated with the transport of planetary ions downstream into the magnetotail (Barabash et al., 2007; Lundin et al., 1989), whereas plume escape is largely driven by the solar wind convection electric field (Dubinin et al., 2011; Fang et al., 2008, 2010). Additionally, ultra-low-frequency (ULF) waves, which are frequently observed in the Martian foreshock region (Delva & Dubinin, 1998; Dubinin & Fraenz, 2016; Jarvinen et al., 2022), can also modulate ion transport through wave-particle interactions (Eastwood et al., 2005).

The ion escape rate is a fundamental parameter for quantifying atmospheric loss on Mars (Fang et al., 2010; Lillis et al., 2017; Lundin et al., 1989, 2013; Ramstad et al., 2015, 2016; Q. Zhang et al., 2023, 2024). Estimating the escape rate and understanding the factors controlling it have therefore been longstanding research topics in Martian space physics (Brain et al., 2016; Lillis et al., 2015; Manning et al., 2011). Using measurements from Mars Express, Ramstad et al. (2015) estimated heavy ion (O^+, O_2^+ and CO_2^+) escape rates on the order of 10^{24} s^{-1} and showed that the escape rate decreases with solar wind density while increasing with solar wind velocity. Lundin et al. (2013) demonstrated that heavy ion escape strongly depends on solar activity, with escape rates during solar maximum exceeding those during solar minimum by approximately one order of magnitude. Using observations from the Mars Atmosphere and Volatile Evolution (MAVEN), Lillis et al. (2017) further indicated that the ion escape rate also varies with both geographic location on Mars and the solar zenith angles (SZAs) of regions with strong crustal magnetic fields. However, in situ spacecraft measurements are limited to local sampling along the spacecraft trajectory and therefore cannot provide a global view of the Martian plasma environment. Furthermore, unlike near-Earth studies, real-time upstream solar wind monitoring is generally unavailable at Mars, making it difficult to directly relate local plasma observations to upstream conditions.

Global numerical simulations provide an important approach for investigating the large-scale structure of the Martian plasma environment. Among them, multi-fluid magnetohydrodynamic (MHD) models, which treat different ion species as separate fluids while neglecting their kinetic effects, have been widely used (C. Dong et al., 2015; Fang et al., 2024; S. Li et al., 2023; Liu et al., 1999; Y. Ma et al., 2004; Y. J. Ma et al., 2015, 2019; Najib et al., 2011; Regoli et al., 2018; Sakata et al., 2024; Sauer et al., 1998; W. Sun et al., 2023). However, kinetic effects play an important role in governing plasma dynamics around Mars (Barabash et al., 2007; Cowee et al., 2012; Shen et al., 2025). The large ion gyroradius near Mars gives rise to finite-Larmor-radius effects, which can strongly influence ion transport and dynamics (Barabash et al., 2007; Kallio & Jarvinen, 2012; Shen et al., 2026). In addition, kinetic effects are essential for describing the pickup and acceleration of planetary ions by the solar wind (Cowee et al., 2012; Fang et al., 2008).

Hybrid simulations provide a compelling alternative (Brecht, 1997; Brecht & Ferrante, 1991; Holmstrom, 2022; Jarvinen et al., 2022; Kallio & Janhunen, 2002; Kallio et al., 2006). This approach captures ion dynamics while remaining computationally feasible for global simulations. Nevertheless, modeling ionospheric ion sources is challenging because ion densities in the ionosphere are orders of magnitude higher than those in the upstream solar wind (Barabash et al., 2007; Fox, 2009). Consequently, most existing hybrid models (Holmstrom, 2022; Jarvinen et al., 2022; Kallio et al., 2006; Wang et al., 2023; Q. Zhang et al., 2023) prescribe ion emission from the upper ionosphere rather than self-consistent modeling ion production processes, except for Brecht (1997). Moreover, Martian crustal magnetic fields are often simplified or neglected, despite observational evidence showing that crustal magnetic anomalies can substantially influence atmospheric ion escape (Fang et al., 2015, 2017; Nilsson et al., 2011; Ramstad et al., 2016).

To address these limitations, we have developed a new three-dimensional (3-D) global hybrid simulation model for the Martian plasma environment that self-consistently incorporates ionospheric ion production and crustal

magnetic fields. Using this model, we have quantified the planetary ion escape rate and examined its dependence on the interplanetary magnetic field (IMF) cone angle θ, defined as the angle between the solar wind velocity and the IMF direction. Simulations are first performed for $\theta = 57°$ and $\theta = 6°$, corresponding to Parker spiral and small-cone-angle IMF configurations, respectively. A parametric survey over θ is then conducted.

2. Global Hybrid Code: GCPIC-M

Based on the general curvilinear plasma simulation code (GCPIC) framework, we have developed a 3-D global hybrid model for Mars, referred to as GCPIC-M, where "M" denotes Mars. The GCPIC framework currently incorporates two numerical schemes: a fully kinetic particle-in-cell (PIC) scheme and a hybrid scheme. In the fully PIC approach, both ions and electrons are treated kinetically as particles, and the model has been successfully applied to studies of whistler-mode chorus waves (Chen, Wang, Li, et al., 2025; Lu et al., 2019; Wang et al., 2024) and magnetosonic waves (J. Sun et al., 2023). In the hybrid approach, ions are treated as fully kinetic particles, whereas electrons are modeled as a fluid that maintains quasi-neutrality. This approach has been widely used in simulations of electromagnetic ion cyclotron (EMIC) waves (Chen, Wang, Lin, et al., 2025; Chen et al., 2026). One implementation within the hybrid module is the Auburn global hybrid code in three dimensions (ANGIE3D) model (Lin et al., 2014, 2017), which has been applied to simulations of the solar wind interaction with the terrestrial magnetosphere (X. Li et al., 2025).

In the GCPIC-M model, we focus on low-frequency physical processes satisfying $\omega \leq \Omega_p$ and $k\rho_p \leq 1$, where ω is the wave frequency, k is the wave number, Ω_p is the proton gyrofrequency, and ρ_p is the proton Larmor radius. Under these assumptions, the displacement current can be neglected, and Ampère's law reduces to

$$\nabla \times \boldsymbol{B} = \frac{4\pi}{c}\boldsymbol{J}, \tag{1}$$

where c is the speed of light, $\boldsymbol{B}$ is the magnetic field, and $\boldsymbol{J}$ is the total current density. Faraday's law is given by

$$\nabla \times \boldsymbol{E} = -\frac{1}{c}\frac{\partial \boldsymbol{B}}{\partial t}, \tag{2}$$

where $\boldsymbol{E}$ is the electric field.

The model includes four ion species, H^+, O^+, O_2^+, and CO_2^+, each treated as an independent kinetic particle population. The ion dynamics are governed by

$$\frac{d\boldsymbol{r}_i}{dt} = \boldsymbol{v}_i, \tag{3}$$

and

$$m_i\frac{d\boldsymbol{v}_i}{dt} = q_i\left(\boldsymbol{E} + \frac{\boldsymbol{v}_i \times \boldsymbol{B}}{c}\right) - m_i\nu(\boldsymbol{u}_i - \boldsymbol{u}_e) - \frac{GM_M m_i}{r_i^3}\boldsymbol{r}_i, \tag{4}$$

where $i =$ H^+, O^+, O_2^+, and CO_2^+. Here, $\boldsymbol{r}_i$, $\boldsymbol{v}_i$, and $\boldsymbol{u}_i$ denote the particle position, velocity and bulk velocity for species i, respectively, and m_i and q_i are the corresponding ion mass and charge. Here, $\boldsymbol{u}_e$ is the electron bulk velocity, ν is an ad hoc current-dependent collision frequency used to model anomalous resistivity, G is the gravitational constant, and M_M is the mass of Mars.

In the multi-ion species plasma, the total current density is defined as

$$\boldsymbol{J} = en_e(\boldsymbol{u}_{\text{ion}} - \boldsymbol{u}_e), \tag{5}$$

where

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$$\boldsymbol{u}_{\text{ion}} = \frac{\sum q_i n_i \boldsymbol{u}_i}{e n_e} \tag{6}$$

is the charge-weighted bulk velocity of all ion species, e is the elementary charge, and n_i and n_e are the ion and electron number densities, respectively. Assuming quasi-neutrality and singly charged ion species, the electron number density satisfies

$$n_e = \sum n_i. \tag{7}$$

Retaining the electron inertial term, $m_e d\boldsymbol{u}_e/dt$, as a correction to the commonly used massless electron momentum equation, the electric field can be written as

$$\boldsymbol{E} = -\frac{\boldsymbol{u}_e \times \boldsymbol{B}}{c} - \frac{\nabla p_e}{e n_e} - \frac{m_e \nu}{e}(\boldsymbol{u}_e - \boldsymbol{u}_{\text{ion}}) - \frac{m_e}{e n_e}\frac{d\boldsymbol{u}_e}{dt}, \tag{8}$$

where m_e and p_e are the electron mass and pressure, respectively. Substituting Equation 8 into Equation 2, the magnetic field is advanced to time using Faraday's law. Further details of the numerical implementation are provided by Swift (1996).

A self-consistent ionospheric ion production module is incorporated, in which ion densities are determined from photochemical equilibrium. The ion species included are H_p^+ (where the subscript "p" denotes planetary origin), O^+, O_2^+, and CO_2^+. Their densities are regulated by photoionization (Rees, 1989; Schunk & Nagy, 2000), impact ionization (Cravens, 1987; Modolo et al., 2016), charge exchange (Modolo et al., 2016; Stancil et al., 1999; Stebbings et al., 1964), and dissociative recombination (Fox & Sung, 2001; Schunk & Nagy, 2000). In particular, the photoionization rates depend strongly on solar extreme ultraviolet (EUV) radiation. To account for this effect, the photoionization coefficients are calculated as functions of altitude and SZA following Rees (1989), using the solar EUV flux model of Richards et al. (1994), in which the F10.7 index is adopted to distinguish between solar maximum and solar minimum conditions. The ionospheric density model employed here is identical to that used in the MHD model of Y. Ma et al. (2004), with detailed chemical reactions listed in Table 1 of W. Sun et al. (2023). Note that ion loss processes in the ionosphere (i.e., $dn_i/dt < 0$), including charge exchange and dissociative recombination, are implemented by reducing the weights of the corresponding particles according to the loss rates, while ion production is represented by increasing the number of simulation particles.

The neutral species included are H, O, O_2, and CO_2. The H density profile is adopted from Chaufray et al. (2015), while the density profiles of O, O_2, and CO_2, together with the associated reaction rates, are taken from Y. Ma et al. (2004). These neutral atmospheres are treated as steady-state background profiles that vary with altitude and solar radiation conditions but remain time independent throughout the simulation because their temporal variability occurs primarily on diurnal and seasonal timescales (Chaufray et al., 2015; Elrod et al., 2017). The Martian crustal magnetic field is incorporated using the 60-order spherical harmonic expansion model of Arkani-Hamed (2001), which provides enhanced spatial resolution of crustal magnetic structures while retaining reliable large-scale field characteristics (Arkani-Hamed, 2004; Y. Ma et al., 2002).

The solar wind enters the simulation domain through the upstream boundary, where solar wind protons H_{SW}^+ are continuously injected. Solar wind and planetary electrons are treated as separate populations, with the solar wind electrons assumed to be adiabatic and the planetary electrons treated as isothermal. The initial IMF is added to the crustal magnetic fields, ensuring $\nabla \cdot \boldsymbol{B} = 0$. The subsolar point is located at 180° longitude and 0° latitude, corresponding to the region of the strongest crustal magnetic anomalies on Mars.

The simulations are performed in the Mars Solar Orbital (MSO) coordinate system, in which the X-axis points from the center of Mars to the Sun, the Z-axis points toward the north of Mars' orbital plane, and the Y-axis completes the right-handed system. The solar wind is assumed to flow along the $-X$ direction. The simulation domain extends from -12 to 6 R_M in the X and Y directions and from -6 to 6 R_M in the Z direction, where $R_M = 3{,}396$ km is the radius of Mars. An inner spherical boundary is placed at an altitude of 100 km above the Martian surface, corresponding to 1.03 R_M from the planetary center. Open boundary conditions are applied at the

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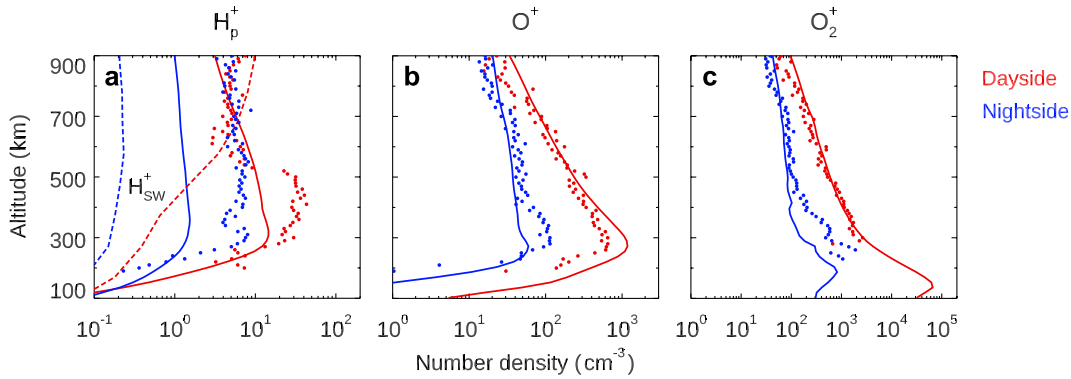

Figure 1. Model-observation comparison of ionospheric ion densities. Altitude profiles of simulated densities for (a) H_p^+, (b) O^+, and (c) O_2^+ shown by solid curves. Dots denote mean values derived from MAVEN/STATIC observations under comparable solar wind conditions. In panel (a), dashed curves represent the density of H_{SW}^+. Red and blue denote averages over the dayside (SZA < 90°) and nightside (SZA > 90°), respectively.

outer boundary, while absorbing boundary conditions are imposed at the inner boundary for both particles and electromagnetic fields.

To better resolve ion production and loss processes in the Martian ionosphere, spherical coordinates with an enhanced radial resolution of 30 km are employed between 1.03 and 2 R_M. Beyond 2 R_M, Cartesian coordinates are adopted to improve computational efficiency in the large-scale solar wind interaction region. The Cartesian grid is nonuniform, with a minimum grid spacing of 0.03 $R_M \times 0.035$ $R_M \times 0.035 R_M$. Because ion production and loss continuously modify the plasma population, the total particle number varies throughout the simulation. As a reference, the smallest grid cell in the upstream solar wind initially contains ~25 particles. The total particle number is $\sim 10^9$ for each planetary ion species and $\sim 10^{10}$ for solar wind protons.

The upstream solar wind protons have a number density of 4 cm^{-3}, a temperature of 5×10^5 K, and a bulk velocity of $\boldsymbol{U}_{sw} = (-400, 0, 0)$ km/s. The upstream electron temperature is set to 2×10^5 K. The IMF magnitude is set to 3 nT, corresponding to an Alfvén Mach number of 12.2. Solar maximum conditions are assumed in the present study. In the ionosphere, the electron temperature follows the altitude-dependent profile derived from the statistical observations of Andrews et al. (2023), increasing from ~100 K at 100 km to $\sim 10^4$ K (~1 eV) above 1,000 km. The ionospheric ion species are initialized with Maxwellian velocity distributions, with ion temperatures set below the local electron temperature but of the same order of magnitude (Hanley et al., 2021; Hanson & Mantas, 1988).

3. Simulation Results

3.1. Parker Spiral IMF Case With $\theta = 57°$

We first perform the Parker spiral case with an IMF cone angle of $\theta = 57°$ and validate the simulated ionospheric ion densities through comparison with MAVEN observations. Figure 1 shows the simulated density profiles as a function of altitude under solar maximum conditions for (a) H_p^+, (b) O^+, and (c) O_2^+, all represented by solid curves. For reference, the simulated density profile of H_{SW}^+ is also included in Figure 1a as dashed curves. The red and blue curves represent averages over the dayside and nightside, defined by solar zenith angles of SZA < 90° and SZA > 90°, respectively. On the dayside, O_2^+ is the dominant ion species, exhibiting a peak density of $\sim 10^5$ cm^{-3} at an altitude of ~150 km. In comparison, O^+ reaches a peak density of $\sim 10^3$ cm^{-3} near 250 km, while H_p^+ attains a maximum density of ~10 cm^{-3} at ~300 km. In addition, the density of H_p^+ is higher than that of H_{SW}^+ at low altitudes below ~700 km.

As expected, the ion densities on the nightside are systematically lower than those on the dayside. This reduction primarily reflects the absence of direct solar illumination, which leads to significantly diminished photoionization (Fowler et al., 2015; Lillis et al., 2021). Moreover, in the absence of sufficient plasma replenishment, chemical loss via recombination further reduces ion densities (Fowler et al., 2015; Lillis et al., 2009).

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The MAVEN spacecraft provides in situ measurements of ionospheric ion composition and densities using the SupraThermal And Thermal Ion Composition (STATIC) instrument (McFadden et al., 2015). Owing to its highly elliptical orbit, MAVEN can sample both the upstream solar wind and the ionosphere within a single orbit (Jakosky et al., 2015). To enable a consistent comparison with the simulation, we select time intervals during which the upstream solar wind, IMF, and solar irradiance conditions measured by MAVEN are comparable to those adopted in the model.

Upstream solar wind parameters are derived from intervals identified as pristine solar wind using the region classification algorithm developed by Halekas et al. (2017). Based on this method, a data set of orbit-averaged upstream solar wind parameters is constructed. Solar irradiance is characterized using the EUV flux measured by the MAVEN EUV Monitor (EUVM; Eparvier et al., 2015), specifically Channel A (17–22 nm), following Y. Dong et al. (2017). The selection criteria are defined as follows: a solar wind proton density of 3–5.5 cm^{-3}, proton temperature below 2×10^5 K, solar wind velocity in the X direction between −450 and −350 km/s, IMF strength of 2–4 nT, cone angles in the range of 45°–70°, and EUV flux exceeding 0.25 mW/m^2. For reference, the multi-year average EUV flux during 2016–2023 (excluding solar maximum years) is approximately 0.21 mW/m^2. Based on these criteria, a total of 50 orbital intervals were selected from MAVEN observations from 2015 to 2024.

The dots in Figure 1 represent ion density profiles measured by MAVEN and averaged over the selected orbital intervals, with the dayside and nightside treated separately. The simulated O^+ and O_2^+ densities agree well with the MAVEN measurements in both magnitudes and altitude-dependent variations on both the dayside and nightside. The STATIC instrument cannot distinguish H_p^+ and H_{SW}^+ and thus measures their combined density. On the dayside, the simulated and observed proton densities are comparable below ∼300 km and above ∼500 km, whereas the simulated densities are lower by a factor of several between ∼300 and ∼500 km. This discrepancy may partly arise from MAVEN sampling of magnetosheath plasma, which can extend to ∼400–500 km near the subsolar region (Dubinin et al., 2008; Vaisberg et al., 2018). On the nightside, the simulated and observed proton densities agree reasonably well below ∼200 km, but the simulated proton density is nearly an order of magnitude lower at higher altitudes. This discrepancy may reflect uncertainties in ion transport and loss processes, as well as sensitivity of the nightside plasma distribution to the initial conditions and simulation duration. The underlying causes of the dayside and nightside discrepancies require further investigation.

The selected intervals include only sparse measurements of O^+ and O_2^+ below ∼200 km, and observations of CO_2^+ are extremely limited; therefore, CO_2^+ is not included in Figure 1. To further assess the model performance, we also conducted a validation case using solar minimum conditions. The simulated CO_2^+ densities, together with the low-altitude O^+ and O_2^+ profiles, agree well with the Viking measurements (Hanson et al., 1977; Y. Ma et al., 2004), providing additional confidence in the modeled ionospheric densities. To avoid confusion with the solar-maximum cases discussed elsewhere in this study, this solar-minimum comparison is presented separately as Figure A1 in the Appendix A.

Figure 2 shows the spatial distributions of key plasma quantities: (a) the total magnetic field strength $B_t = \sqrt{B_x^2 + B_y^2 + B_z^2}$, (b) the B_z component, (c) the solar wind proton density H_{SW}^+, and (d) the planetary proton density H_p^+ in the X–Z plane at $Y = 0$, the X–Y plane at $Z = 0$, and the Y–Z plane at $X = 0$. The arrows in Figure 2a indicate the directions of the solar wind bulk velocity $\boldsymbol{U}_{sw}$, the IMF $\boldsymbol{B}_{sw}$, and the solar wind convection electric field $\boldsymbol{E}_{sw} = -\boldsymbol{U}_{sw} \times \boldsymbol{B}_{sw}$. The bow shock, generated by the solar wind-Mars interaction, is well reproduced in the simulation. Its subsolar standoff distance is ∼1.6 R_M from the center of Mars, and its cross section in the $X = 0$ plane is nearly circular, with a radius of ∼1.6 R_M. The overall bow shock location is consistent with the empirical model (Vignes et al., 2000), as indicated by the solid curves. The dynamic evolution of the magnetic field strength is presented in Movie S1.

Downstream of the bow shock lies the magnetosheath, followed by the IMB. The IMB has a subsolar standoff distance of ∼1.3 R_M (dashed curves) and appears approximately circular in the $X = 0$ plane with a radius of ∼1.4 R_M. Inside the IMB, the magnetic field is enhanced by IMF pileup and draping, while crustal magnetic fields can make substantial contributions, particularly at lower altitudes and in regions of strong crustal magnetization. The IMB is less evident in the linearly scaled B_t distribution, but becomes more discernible under logarithmic scaling

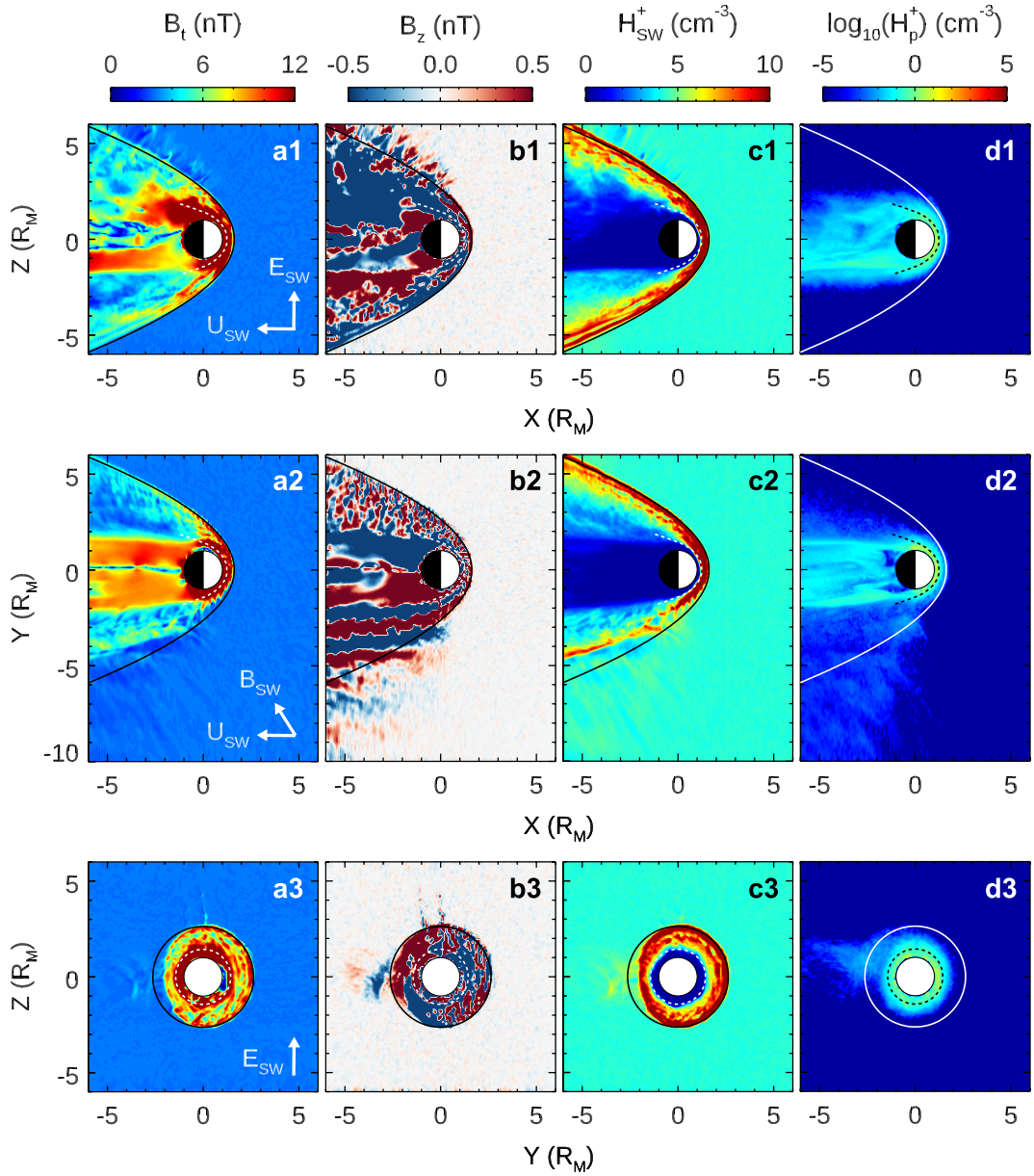

Figure 2. Overview of the Parker spiral case with $\theta = 57°$ at $t = 17.11$ min. Physical quantities in the (a1–d1) X–Z ($Y = 0$), (a2–d2) X–Y ($Z = 0$), and (a3–d3) Y–Z ($X = 0$) planes. From left to right: total magnetic field B_t, the B_z component, solar wind proton density H^+_{SW}, and the logarithmic number density of H^+_p. Solid and dashed curves (black or white) denote the empirical bow shock and IMB locations, respectively, based on Vignes et al. (2000). Arrows in panels a1-a3 indicate the directions of $\boldsymbol{B}_{SW}$, $\boldsymbol{U}_{SW}$, and $\boldsymbol{E}_{SW}$.

(not shown). It can be more clearly identified in the proton density distribution (Figures 2c and 2d) and in the heavy ion density distributions shown in Figure 3.

As shown in Figures 2a2–2d2, the bow shock is asymmetric with respect to $Y = 0$, consistent with satellite observations (Cheng et al., 2024; Edberg et al., 2025). On the $Y < 0$ side, corresponding to the quasi-parallel shock, a proton foreshock develops upstream of the bow shock. This region is characterized by fluctuations in B_t and B_z, as well as in the densities of H^+_{SW} and H^+_p. These perturbations are associated with upstream ULF waves generated by backstreaming protons (Jarvinen et al., 2022; Le & Russell, 1996; Tanaka et al., 1983), including both reflected solar wind protons and escaping planetary protons. However, the proton density fluctuation ($\sim$1 cm^{-3}) significantly exceeds the planetary H^+_p density, indicating that the foreshock dynamics are primarily

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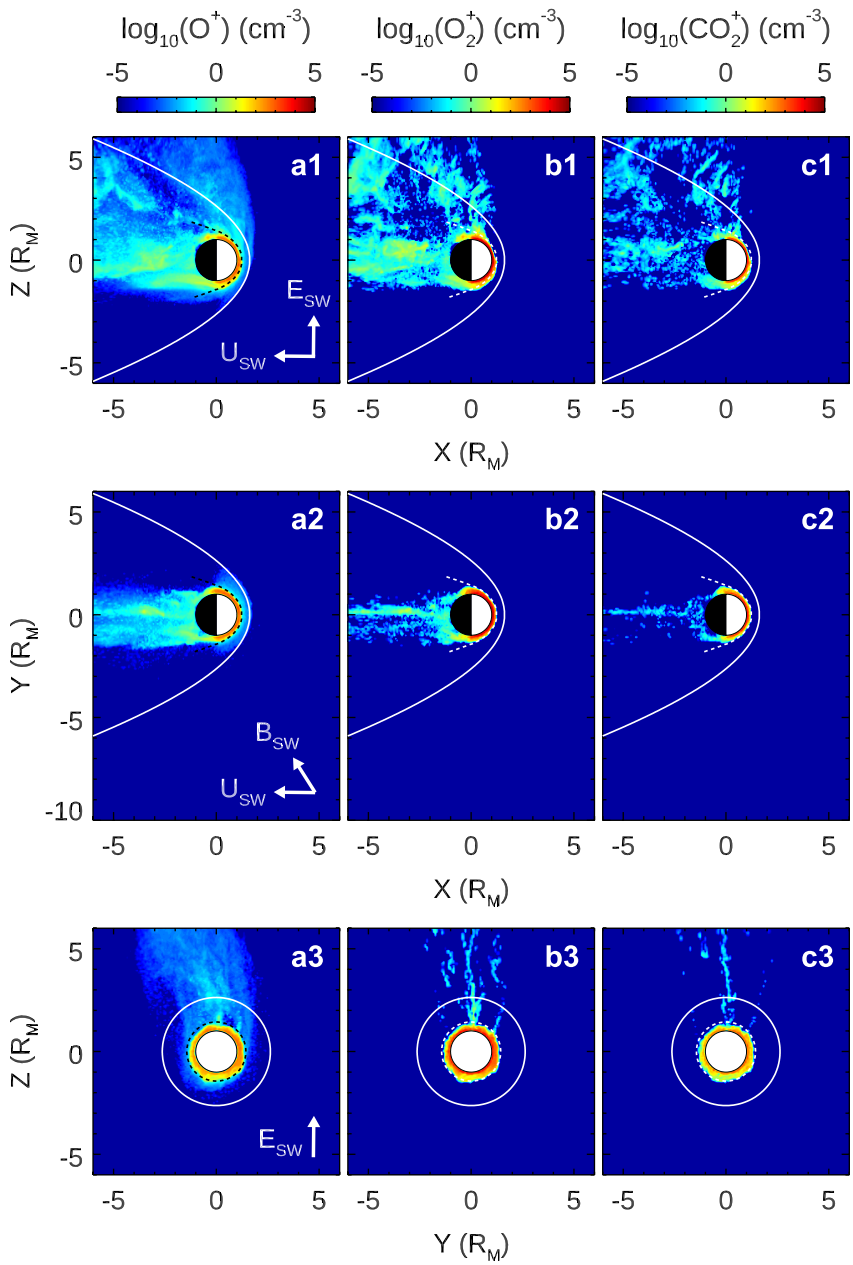

Figure 3. Overview of the heavy ion densities in the Parker spiral IMF case. The logarithmic number density of O^+, O_2^+, and CO_2^+ in the (a1–c1) X–Z ($Y = 0$), (a2–c2) X–Y ($Z = 0$), and (a3–c3) Y–Z ($X = 0$) planes. Solid and dashed curves (black or white) denote the empirical bow shock and IMB locations, respectively. Arrows in panels a1–a3 indicate the directions of $\boldsymbol{B}_{SW}$, $\boldsymbol{U}_{SW}$, and $\boldsymbol{E}_{SW}$.

controlled by the reflected solar wind protons rather than the planetary protons. The presence of the foreshock modifies the local shock structure, with reduced magnetic field strength and weakened proton density compression in this region, leading to a local inward deformation of the shock surface. Similar magnetic field and density depletions have been found in ion foreshock regions at Mercury (Jarvinen et al., 2020), where the intrinsic magnetic field is also weak.

Although a fraction of H_p^+ exhibits outflow in the –Y direction in the foreshock region, the dominant transport remains along the –X direction (Figures 2d1 and 2d2), which is attributed to the Hall electric field $\boldsymbol{J} \times \boldsymbol{B}$ (S. Li et al., 2023; C. Zhang et al., 2022). This preferential transport reflects the magnetized nature of H_p^+, which tend to follow magnetic field lines under the frozen-in condition. The evolution of the H_p^+ density is presented in

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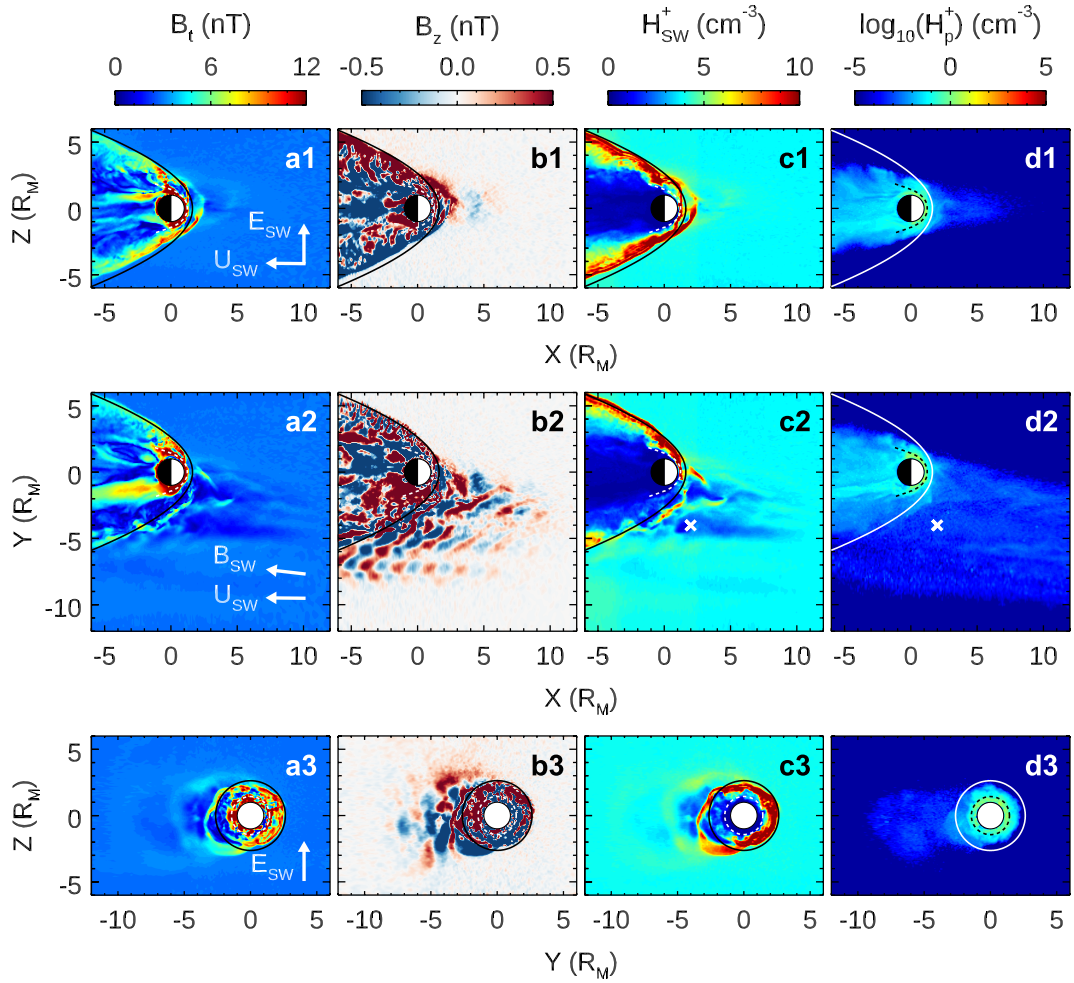

Figure 4. Overview of the small-cone-angle IMF case with $\theta = 6°$ at $t = 17.11$ min. Same format as Figure 2, but shown over an extended range along the X and Y directions. In panels (c2) and (d2), the white crosses mark the position of (2, −4, 0) R_M.

Movie S2. To quantify ion escape, we calculated the number of planetary ions crossing each outer boundary surface per unit time, with the escape in different directions evaluated separately. Only ions crossing the outer boundaries are counted as escaping; ionospheric ion losses are not included in the escape rate. The escape rates are evaluated after the system reaches a quasi-steady state, when they vary only weakly with time. For H_p^+, the escape rates are 1.3×10^{24} s^{-1} through the $X = -12\ R_M$ boundary (–X direction; tailward escape) and 4.4×10^{22} s^{-1} through the $Y = -12\ R_M$ boundary (–Y direction).

Figure 3 presents the spatial distributions of heavy ion densities, including O^+, O_2^+, and CO_2^+. Both the heavy ions and H_p^+ exhibit peak densities within the ionosphere, demonstrating the importance of incorporating a self-consistent ionospheric ion production model. In contrast to H_p^+, however, heavy ions display not only tailward escape but also a pronounced plume escape through the $Z = 6\ R_M$ boundary (+Z direction). For example, the two escape channels of O^+ are shown in Movie S3. As shown in Figures 3a1–3c1 and 3a3–3c3, these ions form plume-like density distributions that are largely decoupled from the bow shock, owing to their large gyroradii. These demagnetized ions are pickup ions (PUIs) originating from the Martian atmosphere and are distributed from the shock front into the downstream region (Curry et al., 2015; Dubinin et al., 2011; Fang et al., 2008; Luhmann et al., 1992), moving along the convection electric field $\boldsymbol{E}_{SW}$. The escape rates for O^+, O_2^+, and CO_2^+ are 3.1×10^{23}, 4.2×10^{23}, and 7.8×10^{22} s^{-1} along the $-X$ direction, and 2.6×10^{23}, 5.0×10^{23}, and 8.5×10^{22} s^{-1} along the $+Z$ direction, respectively. The total heavy ion escape rate is 1.6×10^{24} s^{-1}, comparable to the MHD simulation estimate of 1.7×10^{24} s^{-1} obtained by Y. Ma et al. (2004) under the same upstream solar wind

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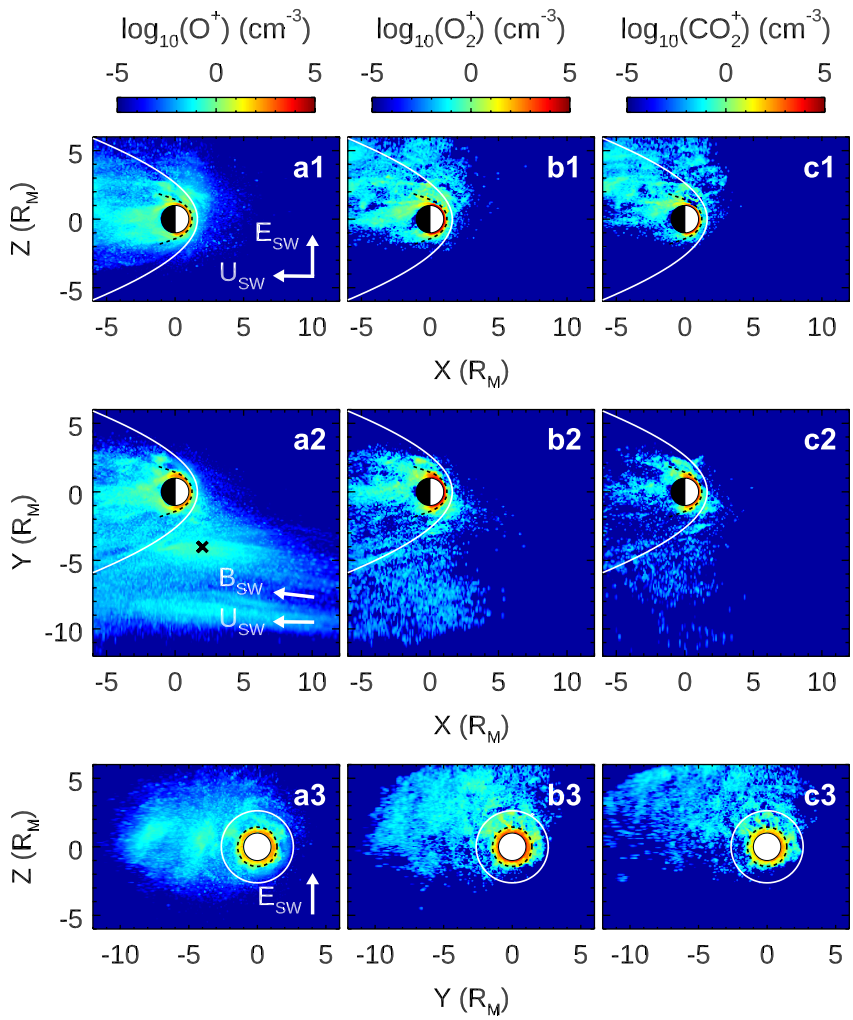

Figure 5. Overview of the heavy ion densities in the $\theta = 6°$ case. Same format as Figure 3, but shown over an extended range along the X and Y directions. The black cross in panel (a2) marks the position of (2, −4, 0) R_M.

conditions. Among the three heavy ion species, O^+ exhibits the most spatially extended distribution. In the X–Z plane, enhanced O^+ densities are preferentially observed in the $+Z$ hemisphere, forming a broad plume-like structure that extends several Martian radii away from the planet.

3.2. Small-Cone-Angle IMF Case With $\theta = 6°$

We then examine a small-cone-angle IMF case with $\theta = 6°$, also referred to as a quasi-radial IMF configuration in previous studies (Jarvinen et al., 2022; Lin et al., 2026; Song et al., 2026; Q. Zhang et al., 2024). All other upstream solar wind conditions are kept the same as in the Parker spiral case. Due to the small cone angle, the convection electric field $\boldsymbol{E}_{\rm SW}$ is reduced (Q. Zhang et al., 2024). Figure 4 presents the spatial distributions of (a) B_t, (b) B_z, (c) $H^+_{\rm SW}$, and (d) H^+_p on three orthogonal slices. In the X–Y plane, the quasi-perpendicular shock on the $Y > 0$ side is generally consistent with the empirical model. In contrast, near the subsolar region and along the quasi-parallel shock on the $Y < 0$ side, strong foreshock wave activity drives the highly dynamic evolution of the bow-shock structure. The shock front undergoes rapid formation and restructuring, making a well-defined shock surface difficult to identify in an instantaneous snapshot. This dynamic evolution can be seen in Movie S4. In the X–Z plane, the simulated bow shock generally follows the empirical model, although the agreement is less clear near the subsolar region. Compared with the Parker spiral case, this case produces a larger proton foreshock region and stronger transverse wave activity, as shown by the B_z contours. The shock-deformation-associated region also extends farther upstream (Figures 4a–4b), consistent with the observations of Hanley et al. (2026).

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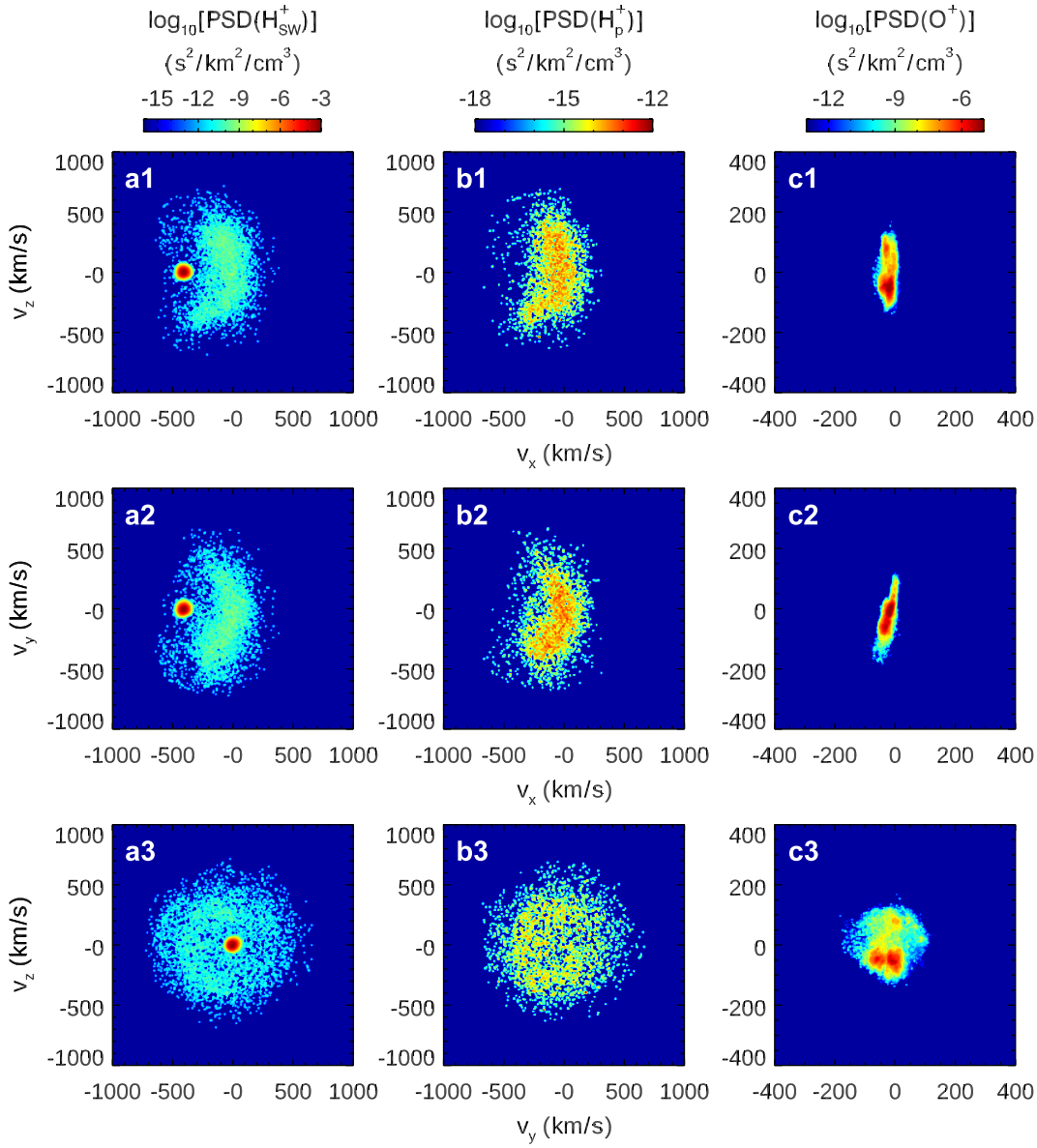

Figure 6. Ion velocity distributions in the proton foreshock. PSDs of (a) H^+_{SW}, (b) H^+_p, and (c) O^+ at (2, − 4, 0) R_M and 17.11 min. Rows 1–3 correspond to the v_x–v_z, v_x–v_y, and v_y–v_z planes, respectively.

Concurrently, both H^+_{SW} and H^+_p exhibit density perturbations extending beyond 12 R_M in the $+X$ direction, indicating field-aligned streaming of protons of both solar wind and planetary origin into the upstream. Correspondingly, the escape channel of H^+_p exhibits a directional shift: ion outflow that was previously concentrated in the $-Y$ direction is redirected toward the $+X$ direction (Movie S5). Nevertheless, tailward escape remains the dominant loss channel. Specifically, the H^+_p escape rates are estimated to be 1.2×10^{24} s^{-1} in the tailward direction and 2.3×10^{22} s^{-1} in the $+X$ direction. The upstream ion motion could be partly attributed to the $\boldsymbol{E} \times \boldsymbol{B}$ drift driven by the convective electric field (Q. Zhang et al., 2024).

For a small IMF cone angle, the reduction in $\boldsymbol{E}_{SW}$ also causes heavy ions to deviate from the typical plume escape trajectories and exhibit pronounced gyromotion (Figure 5). Taking O^+ as an example, these ions complete approximately two gyro-orbits in the simulation domain (Figures 5a2–5a3; Movie S6), leading to gyrophase bunching (GPB; Mazelle et al., 2003) in the X–Y plane. Gyromotion is also evident for O^+_2 and CO^+_2 (Figures 5b3 and 5c3), with larger gyroradii than O^+ due to the larger mass.

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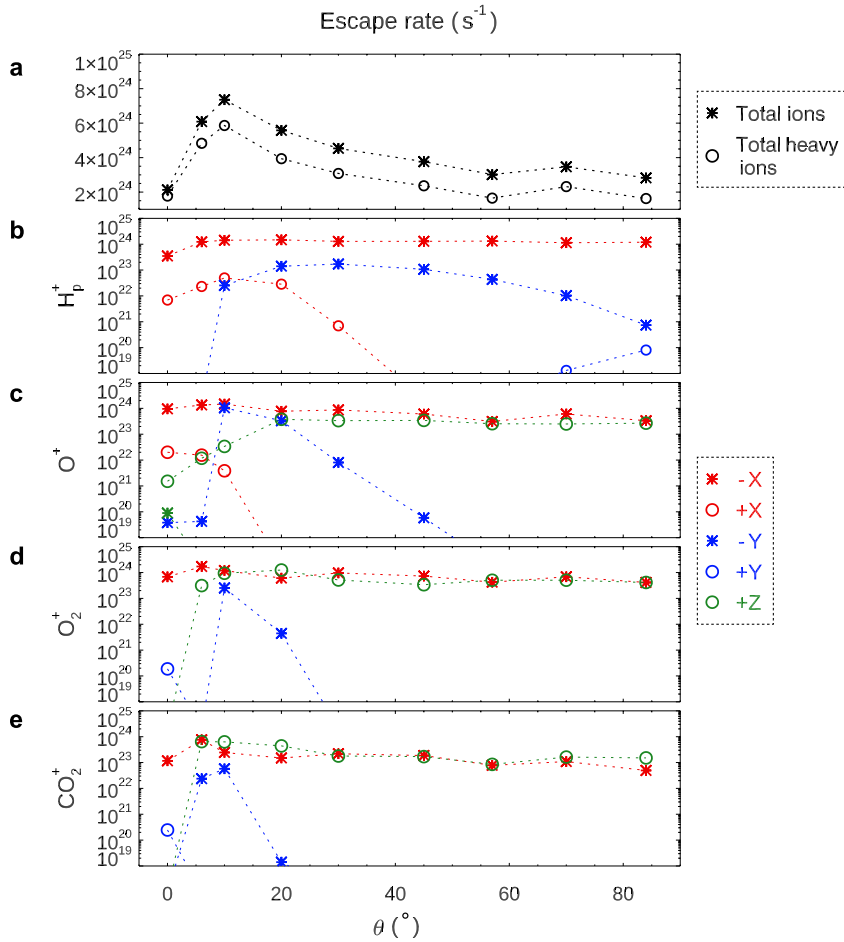

Figure 7. Estimation of escape rates. (a) Total ion escape rate (black asterisks) and heavy ion escape rate (black circles) as a function of IMF cone angle θ. Escape rates for (b) H_p^+, (c) O^+, (d) O_2^+, and (e) CO_2^+. In panels (b–e), red asterisks represent the $-X$ (tailward) direction, and red circles represent the $+X$ direction; blue asterisks represent $-Y$ direction, and blue circles represent the $+Y$ direction; green circles represent the $+Z$ (plume) direction.

The O^+ escape is dominated by the tailward direction, with a rate of 1.4×10^{24} s^{-1}, while plume escape is much weaker at 1.2×10^{22} s^{-1}. The foreshock and its associated waves introduce additional escape channels, including a $+X$-directed component with a rate of 1.6×10^{22} s^{-1} and a weaker $-Y$-directed component at 4.2×10^{19} s^{-1}. For O_2^+ and CO_2^+, the tailward and plume escape rates are comparable, with values of 1.7×10^{24} and 3.1×10^{23} s^{-1} for O_2^+, and 7.6×10^{23} and 6.5×10^{23} s^{-1} for CO_2^+, respectively. Moreover, CO_2^+ exhibits a notable $-Y$-directed escape with a rate of 2.4×10^{22} s^{-1}. The simulated total heavy ion escape rate is 5.5×10^{24} s^{-1}, higher than the MHD estimate of 2.8×10^{24} s^{-1} using the same upstream conditions (Y. Ma et al., 2004), due to the inclusion of foreshock kinetics in the hybrid simulation. For even smaller IMF cone angles, the convection electric field becomes further reduced and vanishes in the limiting case of $\theta = 0°$. Consequently, heavy-ion escape driven by the convection electric field is strongly suppressed, and the escape is primarily controlled by foreshock wave activity.

To further investigate ion dynamics in the foreshock, ion phase space densities (PSDs) are analyzed in the $\theta = 6°$ case. Figure 6 shows the PSDs in the v_x–v_z, v_x–v_y, and v_y–v_z planes for H_{SW}^+, H_p^+, and O^+ in a local region centered at $(x_0, y_0, z_0) = (2, -4, 0)\ R_M$, marked by crosses in Figures 4c2, 4d2, and 5a2. The two-dimensional PSDs are obtained by spatially averaging the distribution function over this region and integrating over the out-of-plane velocity component:

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$$\mathrm{PSD}_{v_x-v_z} = \frac{1}{V}\int_{x_0-\Delta x/2}^{x_0+\Delta x/2}\int_{y_0-\Delta y/2}^{y_0+\Delta y/2}\int_{z_0-\Delta z/2}^{z_0+\Delta z/2}\int_{-\infty}^{+\infty} f\left(x,y,z,v_x,v_y,v_z\right)\, dv_y\, dz\, dy\, dx, \quad (9)$$

$$\mathrm{PSD}_{v_x-v_y} = \frac{1}{V}\int_{x_0-\Delta x/2}^{x_0+\Delta x/2}\int_{y_0-\Delta y/2}^{y_0+\Delta y/2}\int_{z_0-\Delta z/2}^{z_0+\Delta z/2}\int_{-\infty}^{+\infty} f\left(x,y,z,v_x,v_y,v_z\right)\, dv_z\, dz\, dy\, dx, \quad (10)$$

$$\mathrm{PSD}_{v_y-v_z} = \frac{1}{V}\int_{x_0-\Delta x/2}^{x_0+\Delta x/2}\int_{y_0-\Delta y/2}^{y_0+\Delta y/2}\int_{z_0-\Delta z/2}^{z_0+\Delta z/2}\int_{-\infty}^{+\infty} f\left(x,y,z,v_x,v_y,v_z\right)\, dv_x\, dz\, dy\, dx. \quad (11)$$

Here, $f\left(x,y,z,v_x,v_y,v_z\right)$ is the six-dimensional distribution function of the corresponding ion species, and $V = \Delta x\Delta y\Delta z$ is the integration volume, with $\Delta x = \Delta y = \Delta z = 0.2\ R_M$.

As shown in Figure 6a, the solar wind population appears as a small red spot at (−400, 0, 0) km/s. Reflected protons form a ring distribution in the v_y-v_z plane and are decelerated in the X direction, giving rise to low-energy field-aligned beams (FAB; Thomsen et al., 1983; Schwartz & Burgess, 1984). Notably, planetary protons also form FABs resembling those of solar wind protons (Figure 6b). The similarity between reflected H^+_{SW} and escaping H^+_p is consistent with both populations being scattered by foreshock waves. The thermal component of these protons with $v_x > 0$ can propagate against the solar-wind flow and reach the upstream boundary, contributing to the density contours (Figures 4c2 and 4d2).

The O^+ ions also exhibit field-aligned, nongyrotropic ring-beam features, scattered in the perpendicular (Y-Z) direction. Their velocities in the v_y–v_z plane are lower and form a less thermalized distribution, indicating weaker scattering in the foreshock region. As shown in Figure 6c3, their $+v_z$ acceleration is driven by the convection electric field $\boldsymbol{E}_{\mathrm{SW}}$, while the pronounced $-v_z$ beam component indicates that these heavy ions have completed approximately one gyro-orbit (Figures 5a2 and 5a3).

3.3. IMF Cone Angle Dependence

We further investigate the dependence of ion escape rates on the IMF cone angle θ. In addition to the $\theta = 57°$ and 6° cases, simulations are also performed for $\theta = 0°$, 10°, 20°, 30°, 45°, 70°, and 84°.

Figure 7a shows the total ion escape rate (black asterisks) and the total heavy ion escape rate (black circles). The total escape rate reaches a maximum of 7.4×10^{24} s^{-1} at $\theta = 10°$, and then decreases with increasing θ, gradually approaching a quasi-steady value of $\sim 3 \times 10^{24}$ s^{-1} for $\theta > \sim 57°$. The enhanced escape at relatively small cone angles indicates the important role of foreshock formation in promoting ion escape, highlighting the necessity of including ion kinetic effects in the model. However, the escape rate decreases again for $\theta < 10°$ because the weaker $\boldsymbol{E}_{SW}$ provides less ion acceleration, thereby suppressing plume escape. As θ approaches 0°, the escape rate decreases to $\sim 2 \times 10^{24}$ s^{-1}. The heavy ion escape rate exhibits a similar trend to the total escape rate, with an average value of 2.9×10^{24} s^{-1}, comparable to typical estimates in observations (Lundin et al., 2008; Nilsson et al., 2011; Ramstad et al., 2015).

Figure 7b presents the directional escape rates of H^+_p, whose tailward escape dominates and remains relatively insensitive to θ, with a typical value of $\sim 10^{24}$ s^{-1}. Additional escape channels are identified in the $-Y$ and $+X$ directions, both associated with foreshock wave dynamics. These components peak at small cone angles, reaching 1.7×10^{23} s^{-1} at $\theta = 30°$ ($-Y$) and 5.0×10^{23} s^{-1} at $\theta = 10°$ ($+X$). At large cone angles ($\theta \geq 70°$), a weak $+Y$ escape component ($<10^{20}$ s^{-1}) appears, which is due to the foreshock formation on the $+Y$ side (not shown).

For heavy ions (Figures 7c–7e), tailward escape and plume escape are generally comparable, with rates of $\sim 10^{24}$ s^{-1} for O^+ and O_2^+, and $\sim 10^{23}$ s^{-1} for CO_2^+. At small cone angles, plume escape is also reduced due to the weak $\boldsymbol{E}_{\mathrm{SW}}$. Similar to H_p^+, heavy ions also exhibit $-Y$ escape at small θ, with peak values at $\theta = 10°$.

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4. Discussion

In this study, we mainly focus on the ion foreshock associated with the quasi-parallel bow-shock region on the −Y side. In addition to waves and structures in this region, enhanced wave activity characterized by both B_t and B_z fluctuations is also identified upstream of the bow shock on the +Z side (Figures 2a1 & 2b1). This region corresponds to a quasi-perpendicular shock geometry and is therefore distinct from the ion foreshock. The wave activity there could be associated with agyrotropic heavy-ion outflows driven by $\boldsymbol{E}_{SW}$. The underlying generation mechanism will be investigated in future work. This study represents a first step toward understanding kinetic processes in the Martian plasma environment. Future studies will combine GCPIC-M simulations with MAVEN observations to investigate ion dynamics in different regions and their roles in wave generation and atmospheric escape.

The simulations presented here are performed under solar maximum conditions, where the solar EUV flux is specified using the empirical model of Richards et al. (1994), which parameterizes the EUV intensity through the F10.7 index. Although this approach has been widely adopted in previous studies, the F10.7 index serves only as an indirect proxy of solar EUV activity and cannot fully capture the temporal and spectral variability of the incident solar radiation. The MAVEN EUVM instrument directly measures solar EUV fluxes over different wavelength bands, providing more realistic inputs for ionospheric ion production models. Future implementations of GCPIC-M will incorporate these measured EUV fluxes, enabling more rigorous event-based comparisons between simulations and observations. In addition, low-altitude plasma density measurements from MAVEN can be used to further optimize the H_p^+ and H_{SW}^+ density model and improve the calibration of simulated ionospheric densities. The present simulations do not include the seasonal and diurnal variations of the crustal magnetic field in the MSO frame associated with Mars' orbital motion and rotation. A systematic investigation of these effects on ion escape is left for future work.

5. Summary

We have developed a 3-D global hybrid simulation model for the Martian plasma environment, GCPIC-M, in which ionospheric ions are self-consistently generated through photochemical reactions. Under given upstream solar wind conditions, the simulated heavy ion density profiles in the ionosphere show good agreement with MAVEN observations on both the dayside and nightside. The simulations further demonstrate that planetary protons are transported primarily along the magnetotail, whereas heavy ions exhibit both tailward and plume outflows. The foreshock can also modulate ion transport, leading to enhanced outflows toward the upstream region.

We have further quantified the total ion escape rate and examined its dependence on the IMF cone angle θ. The simulated escape rate is generally on the order of 10^{24} s^{-1}, consistent with typical values in observations. The escape rate reaches a maximum of $\sim 7 \times 10^{24}$ s^{-1} at $\theta \sim 10°$. At larger cone angles, it remains nearly constant at $\sim 3 \times 10^{24}$ s^{-1}, while at smaller cone angles, it decreases substantially, reaching $\sim 2 \times 10^{24}$ s^{-1} as θ approaches 0°, due to the reduction in the solar-wind convection electric field. This nonmonotonic dependence highlights the kinetic nature of Martian ion escape and demonstrates that IMF geometry controls both the total escape rate and the relative contributions of plume, tailward, and foreshock escape channels.

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Appendix A: Comparison of Simulated Low-Altitude Ion Density Profiles With Viking Observations

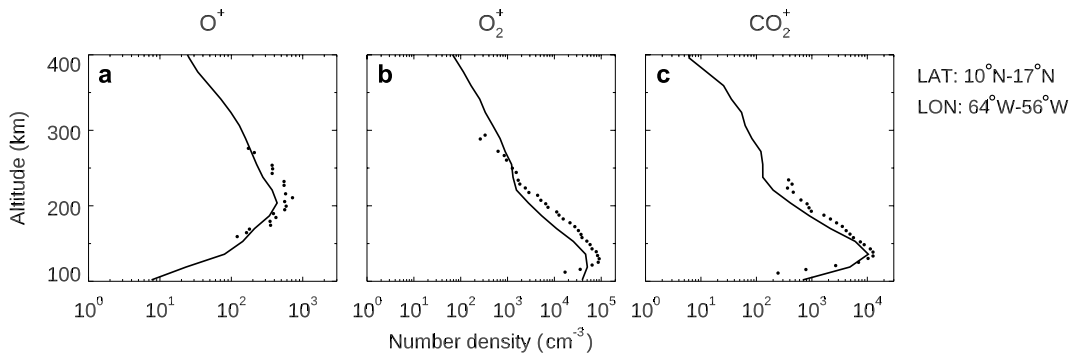

Figure A1. Number density as a function of altitude at latitudes 10°–17°N and longitudes 64°–56°W for (a) O^+, (b) O_2^+, and (c) CO_2^+. Dots denote the Viking measurements from the retarding potential analyzer (RPA; Hanson et al., 1977) under solar minimum conditions. For direct comparison, the simulation is also performed under solar minimum conditions, with all other parameters identical to the Parker spiral IMF case. Details of the Viking observations are described in Y. Ma et al. (2004).

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

The MAVEN data used in this study are publicly available at https://sprg.ssl.berkeley.edu/data/maven/data/sci/. The simulation data generated in this study are available from Wang and Chen (2026).

Acknowledgments
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